via Roigo 14, Cadro (Switzerland)
* Varese, 20.01.1998
+41 76 452 2748
samu9.ferri@gmail.com
ferrixio.github.io

Samuele Ferri

Research interests

Numerical linear algebra, GLT theory, numerical methods for PDEs

Education

2024–now **PhD in Computer Science and Mathematics of Computation**, *University of Insubria*, Como
Expected defense date: February 2027

2020–2022 **Master degree in mathematics**, *University of Insubria*, Como
Graduation grade: 104/110

2022 **Teaching qualification (Italy)**, *University of Insubria*, Como
24 cfu qualification

2017–2020 **Bachelor degree in mathematics**, *University of Insubria*, Como
Graduation grade: 93/110

2013–2017 **High school diploma**, *Liceo Lugano 1*, Lugano

Pubblications

Braun, E. C. and Bretti, G. and Ferri, S. and Santarelli, M. L. and Semplice, M.. "*Dissolution of carbonate stones caused by CO2 pollutant: an erosion model*". Arxiv preprint (2026)

Ferri, S. and Giraudo, C. and Loi, V. and Kuchta, M. and Serra-Capizzano, S.. "*Spectral analysis of the stiffness matrix sequence in the approximated Stokes equation*". Bol. Soc. Mat. Mex, Volume 32, pp. 102 (2026)

Barakitis, N. and Donatelli, M. and Ferri, S. and Loi, V. and Serra-Capizzano, S. and Sormani, R. L.. "*Blocking structures, approximation, and preconditioning*". Numerical Algorithms, Volume 100, pp. 1893–1927 (2025)

PhD thesis

Title On Preconditioning for Block-structured Matrices

Supervisor Stefano Serra-Capizzano, Matteo Semplice

Description In this thesis we review the spectral analysis and preconditioning strategies for block-structured matrices with applications.

Master thesis

Title *De exoticis 7-sphaeris*

Supervisor Giovanni Bazzoni

Description In this paper, we retrace Milnor's studies aimed at finding homeomorphic but not diffeomorphic manifolds on the 7-sphere, through the study of 3-sphere bundles on the 4-sphere and their characteristic classes.

Bachelor thesis

Title	*Mixing time for Glauber dynamics*
Supervisor	Andrea Posilicano
Description	This thesis work is aimed at studying the time required for the Markov chain describing the Glauber dynamics for the Ising model to reach its stationary distribution.

Skills

Programming	Python, Latex, C++, Matlab, HTML, CSS, SQL, LabView, Affinity
Language	Italian (mother t.), English (advanced), German (basic), French (basic)

Experiences

2026-now **Tutoring**, *University of Insubria*, Como
Disciplinary tutor for the first year of the bachelor's degree in mathematics and physics.

2026 **Internship**, *Università della Svizzera Italiana*, Lugano
From January to July.

2025-2026 **Tutoring**, *University of Insubria*, Como
Disciplinary tutor for the first year of the bachelor's degree in mathematics and physics.

2025 **Internship**, *Institute for applied mathematics "Mauro Picone"*, National Research Council of Italy, Rome
From January to July.

2024-2025 **Tutoring**, *University of Insubria*, Como
Disciplinary tutor for the first year of the bachelor's degree in mathematics and physics.

2024 **Tutoring**, *University of Insubria*, Como
For the "*Analisi 2*" course of the bachelor's degree in mathematics.

2024 **Tutoring**, *University of Insubria*, Como
For the "*Matematica 2*" course of the bachelor's degree in chemistry.

2023 **Data analyst**, *Tools for Smart Minds*, Castel Mella (BS)
From June to December.

2023 **Substitution**, *Canobbio Middle School*, Canobbio
For mathematics for a first grade.

2023 **Substitution**, *Centro Professionale di Trevano*, Canobbio
Substitute three teachers for mathematics for vocational baccalaureate classes.

2023 **Substitution**, *Liceo Lugano 1*, Lugano
Substitute for two teachers for mathematics and computer science.

2022 **Substitution**, *Canobbio Middle School*, Canobbio
For basic mathematics for a third and a fourth grade.

2021–2022 **Tutoring**, *University of Insubria*, Como
For the "*Geometria 2*" course of the bachelor's degree in mathematics.

November 11, 2026